

# Types of Machine Learning

Machine Learning (ML) is a part of Artificial Intelligence that allows computers to learn from data and make decisions or predictions without being directly programmed for every task. Machine Learning is mainly divided into three types: **Supervised Learning, Unsupervised Learning, and Reinforcement Learning.**

## 1. Supervised Learning

In supervised learning, the model is trained using a dataset where the input data and the correct output are already given. The model learns the relationship between them and uses it to predict the output for new data.

For example, if we give a model information about houses such as area, number of rooms, and location along with their prices, the model can learn to predict the price of a new house.

Some common applications are:

- Email spam detection
- House price prediction
- Disease prediction
- Image classification

Supervised learning is mainly divided into:

- **Classification** – used when the output is a category, such as spam or not spam.
- **Regression** – used when the output is a numerical value, such as predicting house prices.

## 2. Unsupervised Learning

In unsupervised learning, the model is given data without predefined output labels. The model tries to find patterns, groups, or relationships within the data by itself.

For example, a shopping website can group customers based on their purchasing habits without already knowing which group each customer belongs to.

Some common applications are:

- Customer segmentation
- Grouping similar products
- Finding unusual patterns in data
- Recommendation systems

Clustering is one of the most common techniques used in unsupervised learning.

### 3. Reinforcement Learning

Reinforcement learning works differently from the other two types. Here, an agent learns by interacting with an environment. It receives a **reward** for good actions and a **penalty** for unwanted actions. Over time, it learns which actions give better results.

For example, a computer playing a game can learn which moves are useful by receiving rewards when it performs well.

Some applications include:

- Game playing
- Robotics
- Self-driving systems
- Resource management

### Conclusion

The three main types of Machine Learning are **Supervised Learning, Unsupervised Learning, and Reinforcement Learning**. Each type works differently depending on the kind of data and the problem we want to solve. Machine Learning is widely used in many real-world applications and is an important part of Artificial Intelligence.